Internship Final Report

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Major: Computer Science (Artificial Intelligence & Machine Learning)

Internship Duration: April 1st, 2026 – May 1st, 2026

Company: ShadowFox

Domain: Artificial Intelligence & Machine Learning

Mentor: Mr. Hariharan

Coordinator: Mr. Aakash

Objectives

The primary objectives of this internship were:

- To understand core concepts of **Artificial Intelligence and Machine Learning**
- To gain hands-on experience in building **ML models for real-world prediction problems**
- To learn the complete **Machine Learning pipeline** from data preprocessing to deployment
- To develop skills in **supervised learning algorithms (regression models)**
- To deploy machine learning models using a web framework
- To strengthen practical implementation skills in Python for AI/ML applications

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Tasks and Responsibilities

During the internship, I worked on two major AI/ML projects:

- **1. Beginner Level: Housing Price Prediction (AI/ML Model)**

- Worked on a real-world housing dataset
- Performed:
  - Data preprocessing
  - Handling missing values
  - Feature selection
- Conducted **Exploratory Data Analysis (EDA)** to understand feature relationships
- Built and trained **regression models** for predicting housing prices
- Evaluated model performance using:
  - Mean Absolute Error (MAE)
  - Mean Squared Error (MSE)
- Understood how input features influence prediction outputs

🔹 **2. Intermediate Level: Car Price Prediction System (AI/ML + Deployment)**

- Developed a machine learning model to predict car prices
- Applied:
  - Feature engineering
  - Encoding of categorical variables
  - Model training and evaluation
- Created a **web-based prediction system** using:
  - Flask
- Used:
  - Pandas for data handling
  - NumPy for numerical operations
- Implemented system features:

    - User input form (car details)
    - Backend model prediction
    - Display of predicted price
- From your implementation :
    - Model loading using Pickle
    - Feature encoding (Fuel Type, Seller Type, Transmission)
    - Structured input using DataFrame
    - Prediction output in real-time

Learning Outcomes

- **AI/ML Knowledge**
  - Gained strong understanding of:
    - Supervised learning (Regression)
    - Model training and evaluation
    - Feature engineering
- **Practical Implementation**
  - Built complete AI/ML systems:
    - Housing Price Prediction Model
    - Car Price Prediction Web Application
  - Learned how ML models are used in **real-world applications**
- **Technical Skills**
  - Improved proficiency in:
    - Python
    - Machine Learning libraries

    - Data preprocessing techniques

- **Deployment Skills**
  - Learned how to deploy ML models using:
    - Flask
  - Understood integration between frontend input and backend ML model

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Challenges and Solutions

Challenge 1: Data Preprocessing Issues

Solution:
Handled missing and inconsistent data using cleaning techniques and proper formatting

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Challenge 2: Feature Encoding

Solution:
Converted categorical variables into numerical format using encoding methods to make them suitable for ML model.

Challenge 3: Model Performance

Solution:

- Tested multiple regression approaches
- Selected the best-performing model based on evaluation metrics

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Challenge 4: Model Deployment Errors

Solution:

- Faced issues with model loading and input structure
- Resolved by creating a properly structured DataFrame before prediction
- Ensured consistency between training data and input data

Conclusion

This internship in the domain of **Artificial Intelligence and Machine Learning** provided valuable hands-on experience in building and deploying ML models.

Through projects like housing and car price prediction, I gained practical knowledge of:

- Machine learning workflows
- Model training and evaluation
- Real-world AI/ML applications
- Deployment using web technologies

This experience has strengthened my foundation in AI/ML and prepared me for advanced development and research in this field.

Acknowledgments

I express my sincere gratitude to ShadowFox, especially my mentor, Mr. Hariharan, and coordinator, Mr. Aakash, for their guidance and support throughout my internship.

This report reflects the integration of academic knowledge with practical skills gained during the internship, highlighting my journey of learning, growth, and development in the field of Artificial Intelligence & Machine Learning (AI/ML).